EE-133L PROGRAMMING FUNDAMENTALS

Laboratory Manual

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Lab Section: (A)

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Experiment No. 3:
Introduction to Control Statements

Objectives

The objectives of this lab are:

1. To learn the use of if and if-else statements.
2. To explore nested if-else for multiple conditions.
3. To use the conditional operator (?:) for compact decision-making.
4. To learn how to use logical operators (&&, ||, !) in C.
5. To learn the syntax and usage of the switch statement.
6. To compare if-else and switch statements for multiple conditions.

Prerequisites

Following are the prerequisites of this lab are:

- Basic knowledge of writing simple programs in C language.
- Understanding of variables and arithmetic operations in C.

Introduction

Decision Making

In C programming, decision making is required when different actions need to be performed based on certain conditions. Normally, program statements are executed sequentially, but using decision-making statements we can control the flow of execution.

Selection Statements in C

C provides four types of selection structures in the form of statements.

if Statement (Single-Selection Statement)

- Executes a block of code if a condition is true.
- If the condition is false, the statement is skipped.

if-else Statement (Double-Selection Statement)

- Executes one block of code if the condition is true.
- Executes another block of code if the condition is false.

Nested if-else Statements

- Nested if-else statements are used to test multiple conditions by placing if-else statements inside another if or else block.
- They are helpful when decisions depend on multiple conditions.

Switch Statement (Multiple-Selection Statement)

- Allows one block of code to be executed out of many alternatives.
- The selection depends on the value of an expression.

Example: if-Statement

```c
#include <stdio.h>
int main() {
    int grade = 75;

    if (grade >= 60) {
        printf("Passed\n");
    }
    return 0;
}
```

Example: if-else Statement

```c
#include <stdio.h>
int main() {
    int grade = 55;

    if (grade >= 60) {
        printf("Passed\n");
    }
    else {
        printf("Failed\n");
    }
    return 0;
}
```

Conditional Operator (?:)

C provides the conditional operator (?:), which is closely related to the if-else statement.

- It is the only ternary operator in C (works with three operands).
- Together, these operands form a conditional expression.
- The first operand is the condition to be evaluated.
- The second operand is the value (or expression) returned if the condition is true.
- The third operand is the value (or expression) returned if the condition is false.

Example: Using Conditional Operator

```c
#include <stdio.h>
int main() {
    int grade = 55;

    (grade >= 60) ? printf("Passed\n") : printf("Failed\n");

    return 0;
}
```

Note: Use expressions of the same type for the second and third operands of the conditional operator (?:) to avoid subtle errors.

Output

```
muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ gedit output1.c
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ gcc output1.c -o output1
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./output1
Failed
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Example: Nested if-else Statements

```
#include <stdio.h>
int main() {
    int grade;
    char result;
    printf("Enter your grade: ");
    scanf("%d", &grade);

    if (grade >= 90) {
        result = 'A';
    }
    else if (grade >= 80) {
        result = 'B';
    }
    else if (grade >= 70) {
        result = 'C';
    }
    else if (grade >= 60) {
        result = 'D';
    }
    else {
        result = 'F';
    }
    printf("Your grade is: %c\n", result);
    return 0;
}
```

Logical Operators in C

In C, simple conditions such as counter <= 10, total > 1000, or number != sentinel Value are expressed using relational operators (>, <, >=, <=) and equality operators (==, !=). These allow checking only one condition at a time. If multiple conditions are needed, separate statements or nested if/else blocks are required.
To simplify this, C provides logical operators, which allow combining simple conditions into more complex ones:

- **Logical AND (&&)** – True only if *both* conditions are true.
- **Logical OR (||)** – True if *at least one* condition is true.
- **Logical NOT (!)** – Reverses the result of a condition.

All relational, equality, and logical expressions evaluate to **0 (false)** or **1 (true)**, though any nonzero value is treated as true. Operator precedence also affects evaluation order: for example, ! has higher precedence than ==, and both == and >= are evaluated before &&. Logical operators help express multiple conditions compactly and avoid lengthy nested if statements.

Example

```
int main() {
    int marks, age, feePaid;
    marks = 82;
    age = 19;
    feePaid = 0;

    if (marks >= 70 && age >= 18) {
        printf("Eligible based on marks and age.\n");
    }

    if (marks >= 85 || feePaid == 1) {
        printf("Admission confirmed (either high marks OR fee paid).\n");
    }

    if (!(feePaid == 1)) {
        printf("Fee is still pending.\n");
    }

    return 0;
}
```

Compound Statement

The if selection statement expects only one statement in its body, if you have only one statement in the if's body, you do not need to enclose it in braces. To include several statements in the body of an if, you must enclose the set of statements in braces ({ and }). A set of statements contained within a pair of braces is called a compound statement or a block.

Code

```c
#include <stdio.h>

int main() {
    int temperature;

    printf("Enter the temperature in Celsius: ");
    scanf("%d", &temperature);

    if (temperature >= 30) {
        printf("It's a hot day.\n");
    }
    else {
        printf("It's a cool day.\n");
        printf("You might need a jacket.\n");
    }

    return 0;
}
```

Output

```
muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ gedit output2.c
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ gcc output2.c -o output2
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./output2
Enter the temperature in Celsius: 37
It's a hot day.
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Example: Switch Statement

```
#include <stdio.h>

int main() {
    char SHAPE;
    float RADIUS, SIDE, LENGTH, WIDTH, BASE, HEIGHT;

    printf("Enter shape (Circle/Square/Rectangle/Triangle): ");
    scanf(" %c", &SHAPE);

    switch (SHAPE) {
        case 'C':
            printf("Enter radius: ");
            scanf("%f", &RADIUS);
            printf("Area of Circle = %.2f\n", 3.14159 * RADIUS * RADIUS);
            break;
        case 'S':
            printf("Enter side: ");
            scanf("%f", &SIDE);
            printf("Area of Square = %.2f\n", SIDE * SIDE);
            break;
        case 'R':
            printf("Enter length and width: ");
            scanf("%f %f", &LENGTH, &WIDTH);
            printf("Area of Rectangle = %.2f\n", LENGTH * WIDTH);
            break;
        case 'T':
            printf("Enter base and height: ");
            scanf("%f %f", &BASE, &HEIGHT);
            {
                float AREA = 0.5 * BASE * HEIGHT;
                printf("Area of Triangle = %.2f\n", AREA);
            }
            break;
        default:
            printf("Invalid selection.\n");
    }

    return 0;
}
```

Output

```
muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ gcc output3.c -o output3
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./output3
Enter shape (Circle/Square/Rectangle/Triangle): C
Enter radius: 5
Area of Circle = 78.54
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./output3
Enter shape (Circle/Square/Rectangle/Triangle): R
Enter length and width: 4
2
Area of Rectangle = 8.00
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Lab Tasks

Task 1

Write a program which takes three integers from the user and displays the maximum integer of the three.

```
Open  max.c
~/Desktop/PF_Labs/Lab03_2025_EE_016
#include <stdio.h>

int main(){
int a, b, c, max;
printf("Enter 1st Integer: ");
scanf("%d", &a);
printf("Enter 2nd Integer: ");
scanf("%d", &b);
printf("Enter 3rd Integer: ");
scanf("%d", &c);
max = a;
(b > max)? (max = b): (max = a);
(c > max)? (max = c): (max = a);
```

Output

```
muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ make max
gcc -c max.c
gcc max.o -o max
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./max
Enter 1st Integer: 10
Enter 2nd Integer: 100
Enter 3rd Integer: 500
Max: 500
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Task 2

Complete the following code and remove all the errors to get the output as shown below.

Bugged Code

```
printf("Enter your age:");
scanf("%f", age);

if (age >= 18):
{
    printf("You are eligible for the driving license.\n");
}

else:
    print('You are not eligible for the driving license.\n')
}
```

Corrected Code

```
#include <stdio.h>

int main(){
int age; //declared age
printf("Enter your age:");
scanf("%d", &age);      //replaced %f with %d and put &
if (age >= 18)          //removed :
{
printf("You are eligible for the driving license.\n");
}
else{//removed :
printf("You are not eligible for the driving license.\n"); //put ;

}

}
```

Output

```
muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ make license
gcc -c license.c
gcc license.o -o license
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./license
Enter your age:19
You are eligible for the driving license.
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Task 3

Write a C program that takes a year as input from the user and uses nested if to check whether it is a leap year or not.

Code

```
Open ▾   leap_year.c
         ~/Desktop/PF_Labs/Lab03_2025_EE_016
#include <stdio.h>

int main(){
int year;
printf("Enter year: ");
scanf("%d", &year);

if ((year % 4 == 0) && (year % 100 != 0)) {
        printf("Leap Year!\n");
}
else {
        if (year % 400 == 0){
        printf("Leap Year!\n");
        }
        else{
        printf("Not a Leap Year!\n");}

}

}
```

Output

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ make leap_year
gcc -c leap_year.c
gcc leap_year.o -o leap_year
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./leap_year
Enter year: 2026
Not a Leap Year!
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./leap_year
Enter year: 2024
Leap Year!
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```

Task 4

Write a C program that prompts the user to enter an integer number from 1 to 7 and displays the corresponding day of the week. The program should use a **switch statement** to determine the day. If the user enters a number outside this range, the program should display a message indicating **"Invalid input"**.

Code

weekday.c
~/Desktop/PF_Labs/Lab03_2025_EE_016

```c
#include <stdio.h>
int main(){
int day;
printf("Enter the number: ");
scanf("%d", &day);

switch (day){

case 1:
        printf("Monday\n");
        break;
case 2:
        printf("Tuesday\n");
        break;
case 3:
        printf("Wednesday\n");
        break;
case 4:
        printf("Thursday\n");
        break;
case 5:
        printf("Friday\n");
        break;
case 6:
        printf("Saturday\n");
        break;
case 7:
        printf("Sunday\n");
        break;
default:
        fprintf(stderr, "Invalid Number!");
        break;

}
}
```

Output

muhammad-umar@ThinkPad-X395: ~/Desktop/PF_Labs/Lab03_2025_EE_016

```
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ make weekday
gcc -c weekday.c
gcc weekday.o -o weekday
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./weekday
Enter the number: 4
Thursday
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$ ./weekday
Enter the number: 7
Sunday
muhammad-umar@ThinkPad-X395:~/Desktop/PF_Labs/Lab03_2025_EE_016$
```